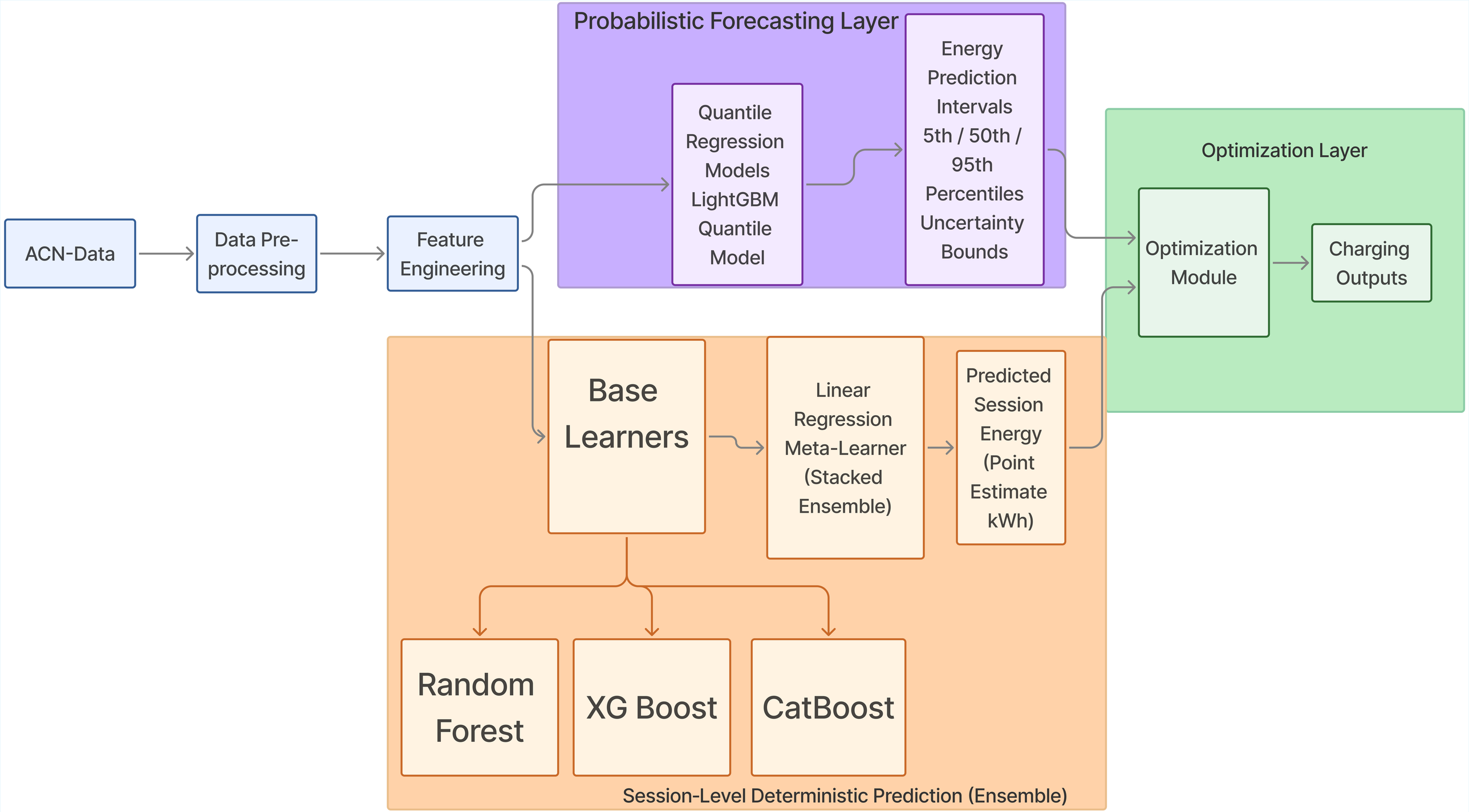
Probabilistic Forecasting Layer
ACN-Data
Data Pre-processing
Feature Engineering
Quantile Regression Models LightGBM Quantile Model
Energy Prediction Intervals 5th / 50th / 95th Percentiles Uncertainty Bounds
Optimization Layer
Optimization Module
Charging Outputs
Base Learners
Linear Regression Meta-Learner (Stacked Ensemble)
Predicted Session Energy (Point Estimate kWh)
Random Forest
XG Boost
CatBoost
Session-Level Deterministic Prediction (Ensemble)